

Presenting a Pseudo-loud Vocal Agency on VR for Stress Reduction

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Abstract

In Japan, mental stress among workers has emerged as a significant social issue due to its harmful effects in both the short and long term. In the short term, stress often leads to problematic behaviors known as “acting out,” which, despite its harmful nature, is easily chosen and even perceived as a convenient stress-relief method. To address this, our study aimed to suppress harmful behavioral activation by inducing harmless alternatives. Specifically, we sought to create a VR-based destructive experience using pseudo-loud vocalization that does not require actual voice production, achieved through auditory stimuli and laryngeal vibratory stimuli. Experimental results showed that vibratory stimuli enhanced the loud vocal agency and may have amplified the positive emotional effects of vocalization itself. Conversely, the visual scenes elicited not only a sense of exhilaration but also feelings of surprise and fear, failing to deliver a generalized stress-relief effect.

CCS Concepts

• Human-centered computing → Haptic devices; Virtual reality.

Keywords

Haptics, Sense of agency, Virtual reality, Vocalization

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1 Introduction

The Ministry of Health, Labour and Welfare indicates that 82.7% of Japanese workers report experiencing anxiety, concerns, or stress related to their work or professional lives [15], identifying this as a significant social issue. Stress exerts a greater psychological impact than is often assumed [29] and is also associated with reduced workplace productivity. While chronic stress is harmful, it can also trigger immediate, uncontrolled behaviors referred to as “acting out” [9, 30]. Such behaviors frequently result in severe negative

consequences for individuals [9]. Nevertheless, certain stress-relief activities, such as “rage rooms,” deliberately incorporate these behaviors [17]. These services allow users to reduce stress by smashing objects, throwing axes, or drawing graffiti on walls. However, there are concerns that these methods may encourage violent and harmful coping mechanisms in daily life [14].

To address this concern, we propose utilizing virtual reality (VR) experiences that simulate behaviors analogous to acting out but in ways that are impossible to replicate in reality. This approach seeks to mitigate the emergence of harmful behaviors under stress while offering a controlled and safe method for stress relief. VR-based methods have demonstrated the ability to naturally and ecologically induce valid emotional responses [26] and are recognized as a promising strategy for stress reduction [1, 7, 12].

Acting out encompasses behaviors such as verbal aggression and physical violence, with actions that directly harm others or constitute criminal behavior being the least ethically acceptable. This study focuses on non-harmful destructive strategies, such as those employed in rage rooms, and tantrum-like strategies that neither harm others nor damage property. However, from the reproducibility perspective, destructive actions involving physical means are undesirable. Consequently, we shifted our attention to voice, one of few non-material human attributes. Specifically, we hypothesized that loud vocalization in a VR setting, such as unrealistically destroying large objects, could serve as a safe, non-harmful method of acting out without encouraging harmful reproducibility. Furthermore, we incorporated pseudo-loud vocalization [21]. This method employs auditory and laryngeal vibratory stimuli to users, substituting for actual loud vocalization. This approach overcomes the physical strain of loud vocalization on the throat and addresses situations where vocalization is difficult, thereby creating a more robust and inclusive experience.

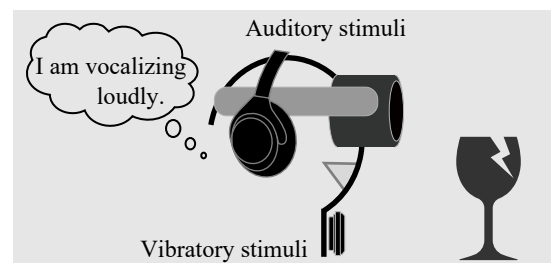


Figure 1: Pseudo-loud vocalization, the concept of this study (adapted from [22], with modifications).

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This study is structured in two stages. The first stage aims to evaluate whether pseudo-loud vocalization can serve as an effective stress-relief strategy and reduce user stress (as shown in Figure 1). The second stage seeks to examine whether employing harmless, content-driven acting-out strategies can prevent users from adopting harmful acting-out behaviors when they face stress in the future. This paper focuses on the first stage, addressing the initial hypothesis.

2 Related work

2.1 Human responses and behavior to stress

Two conceptual frameworks are well-known for systematically organizing human responses to mental stress. The first is the defense mechanism, a concept originating from Sigmund Freud's psychoanalytic theory [9, 37] and later classified by Vaillant into a hierarchy of four developmental levels [19, 31]. Acting out is classified as a lower-level "immature" defense mechanism. Such lower-level mechanisms are non-adaptive and serve only as temporary strategies, often proving harmful in the long term [37]. Consequently, the adoption of higher-order strategies is generally recommended.

The acting out involves expressing emotions under stress in ways that disregard the safety of oneself and others. While often recognized as problematic behavior, Winnicott regarded acting out as a central process for organizing and overcoming difficult emotions [30]. Therefore, efforts to guide acting out into safer forms have potential applications not only in stress management but also as contributions to the field of social welfare.

The second conceptual framework is coping strategies. Coping strategies are defined as "an organizational construct used to encompass the myriad actions individuals use to deal with stressful experiences [23]." Due to their nature, numerous classification methods for coping strategies have been proposed. Skinner et al. [23] argued that the most common and classical classifications (problem-versus emotion-focused coping [6]) lack conceptual clarity, exclusivity, and comprehensiveness as overarching categories. Instead, they identified 13 potential coping families (e.g., "avoidance", "distraction", and "emotion regulation"). Skinner's framework, however, is insufficient for categorizing the VR-based pseudo-loud vocalization in this study. This experience could be seen as distraction, emotion regulation (releasing negativity and seeking exhilaration), or avoidance (immersing in VR to escape the physical world).

2.2 VR-based Interventions aimed at reducing stress

VR-based interventions for stress relief have gained attention due to their ease of use and strong emotional engagement. In problem-focused approaches, practicing tasks in a VR environment prior to actual performance has been proposed as a method to reduce stress and improve emotional regulation during the main task [16, 28]. Emotion-focused approaches aim to achieve stress relief by placing users in calming environments, such as virtual forests [10, 20, 32], biophilic environment [36], or simulated natural wind settings [11]. Although research on active emotion-focused approaches is less developed compared to passive experiences, Kim et al. [14] suggested that active experiences, such as drawing or throwing objects in a VR environment, may offer stronger stress-relief effects than passive

experiences. Our study proposes a stress-relief approach involving active experiences, uniquely targeting the behavioral manifestation of stress.

3 Materials and Methods

3.1 Pseudo loud-vocalization for stress reduction

The concept of pseudo loud-vocalization involves employing auditory and laryngeal vibratory stimuli as substitutes for actual vocalization. This approach allows users to perceive a sense of agency not only over the sound produced but also over the entire vocalization process. Vocalization is one of the few means by which humans can physically affect their surroundings, and it is seldom associated with destructive behaviors in everyday life. By leveraging this distinctive characteristic, we designed an experience that is both unrealistic in nature and yet sensorially attributable to oneself. Furthermore, as a pseudo experience, it reduces the physical strain and constraints associated with real vocalization, providing a safer and universally accessible stress-relief method.

Achieving stress relief through a pseudo-loud vocalization in a VR environment necessitates incorporating a sense of agency. The sense of agency, defined as "The sense that I am the one who is causing or generating an action" [8], is deeply involved in the self-attribution of actions. Experiences that lack a sense of agency fail to elicit a strong emotional response, rendering them unsuitable for the intended stress-relief purposes in this study. The following sections describe the design of audiovisual and haptic stimuli for implementing the pseudo-loud vocalization.

3.1.1 Designing the visual context. To construct an object destruction experience through pseudo-loud vocalization, ensuring the plausibility of the experience is crucial. Plausibility, as described by Mel Slater [24], refers to the sense that events in a VR environment are actually occurring, and it is considered one of the factors contributing to a sense of presence. A key factor that enhances plausibility is coherence [25], which posits that a VR environment should either align with the physical laws of the real world or behave in ways consistent with user expectations. However, vocalization alone is insufficient to produce significant external effects capable of evoking catharsis. Additionally, destruction that is realistically reproducible may unintentionally reinforce undesirable defense mechanisms associated with acting-out. Consequently, the design required for this study incorporates object destruction that, while deviating somewhat from real-world physical laws, remains consistent with user expectations within the VR environment.

Based on the proposed design, we selected a scenario where users vocalize loudly in a classroom setting to destroy a blackboard. The classroom was chosen as it provides an enclosed environment that limits sound wave diffusion, and the blackboard—a thin, rigid, rectangular object—facilitates an easily imaginable destruction scene. As depicted in Figure 2 (A), the blackboard shatters in response to the user's vocalization, with the fragments falling freely under the influence of gravity. To further enhance the perceived impact of the voice, two additional effects were implemented. First, cracks appear on windows made of brittle material, as shown in Figure 2 (B). Second, the blackboard visibly vibrates prior to shattering. These



Figure 2: Visual scene implemented to evoke a sense of exhilaration. (A) represents the scene where the blackboard is destroyed, and (B) illustrates the formation of cracks on the window.

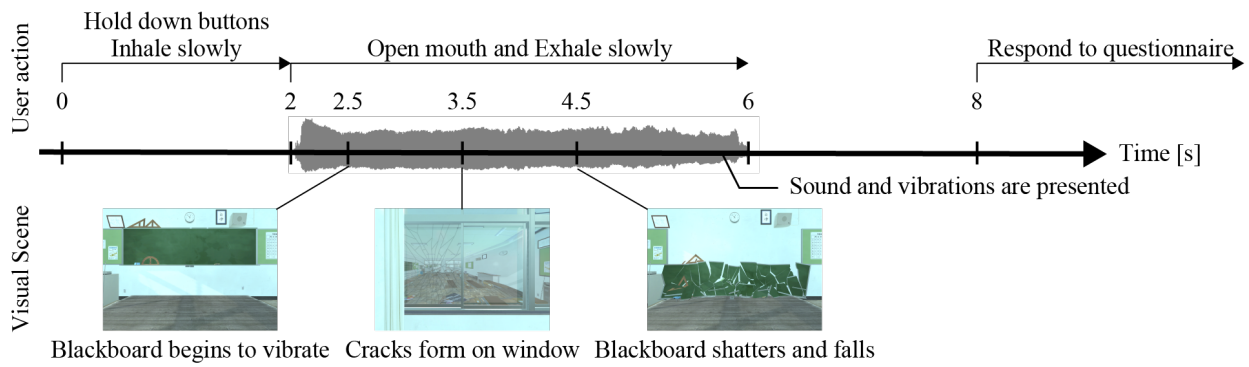


Figure 3: User's VR experience from pressing the button to answering the questionnaire.

effects were designed to amplify the contextual cues of the voice's impact on the external environment, aiming to create a more immersive and convincing pseudo-loud vocalization for users.

3.1.2 Designing the auditory-tactile stimuli. The auditory stimuli consisted of a recorded sound of a male in his 20s vocalizing the vowel /a/ at approximately 80 dB for about 4 s. This volume was selected as it significantly exceeds the typical vocal range of 50 dB to 65 dB observed during normal conversation [4]. The stimuli was calibrated to reach approximately 80 dB at a distance of 0.01 m from the headphone speaker. The vowel /a/ was selected for its suitability in loud vocalizations intended to reduce stress. A male voice was used for the stimuli irrespective of participants' gender to avoid the higher frequencies characteristic of female vocalizations, which might resemble a scream and potentially evoke discomfort or alertness in participants.

For the vibratory stimuli, the recorded audio described earlier was used as-is without applying any special filters, serving directly as the input waveform. The vibration intensity was empirically determined to ensure participant comfort while referencing the intensity levels used in previous studies [21].

3.1.3 Equipment. The experiment was conducted in an experimental area partitioned with dimensions of 1.8 m in width, 2.0 m in length, and 2.7 m in height. Audio was presented using an audio

interface and noise-canceling headphones. Sound levels were measured with a precision sound level meter. For VR visual presentation, a head-mounted display was used, running an environment created and operated using the Unity engine.

To present laryngeal vibratory stimuli, we employed a device based on principles outlined in previous research [21], incorporating a vibro-transducer and an amplifier. Two transducers were symmetrically positioned at the height of the vocal cords, using a band-type device for secure placement. To suppress distortion caused by vibration at the fastening point, as would occur if the Velcro fastener were positioned at the back of the neck, a rubber band was used to encircle the neck seamlessly, eliminating any connection points. The device accepts audio signals as input and produces vibrations on the skin surface that replicate the audio waveform.

3.1.4 Experimental Systems. Participants were seated and equipped with the headmounted display (HMD), as well as the auditory and vibratory stimuli devices, to engage in the VR classroom environment. To initiate the pseudo-loud vocalization, participants pressed and held the middle buttons on both hand controllers simultaneously for more than 2 s. This design was intended to simulate the sensation of effort and buildup associated with actual loud vocalization, even prior to the start of the experience.

Participants were instructed to inhale slowly while holding down the buttons. A gauge appeared simultaneously with the button press, indicating the timing for initiating the pseudo-loud vocalization. Participants were instructed to slightly open their mouths and exhale slowly in alignment with the displayed timing while imagining themselves producing the vocalization of the vowel /a/ loudly. The pseudo-loud vocalization lasted for 4 s. After 0.5 s, the blackboard began to shake, at 1.5 s window glasses shattered, and at 2.5 s the blackboard broke and fell. The trial concluded 6 s after initiation, at which point participants removed the HMD and completed a questionnaire. This sequence of events is illustrated in Figure 3.

3.2 Experimental Design

The research question of this study is whether a pseudo-loud vocalization in VR can reduce stress. The concept of vocal agency in this context is based on the framework proposed by Shimomura et al. [21, 22]. A key distinction of this study from prior research is the inclusion of visual scene with contextual cues that contribute to the sense of agency. These external elements go beyond the classical comparator model of agency [2] and are linked to higher-order judgments of agency [27], suggesting integration with prior information [18]. Therefore, it is hypothesized that inclusion of visual scene will evoke a stronger sense of vocal agency compared to auditory-tactile stimuli alone.

The influence of visual scenes on the sense of vocal agency may depend on laryngeal vibratory stimuli. Vibratory stimuli enhance the sense of embodiment during vocalization and facilitate the attribution of visual outcomes to oneself. Without vibratory stimuli, visual scenes may lead to distinct cognitive effects.

To examine this, a within-participants experiment was conducted with two independent factors: vibration and visual scene. The vibration factor had two levels: “vib-on,” where vibratory stimuli were applied to the larynx in synchronization with the pseudo-loud vocalization while the audio played; and “vib-off,” where no vibratory stimuli were provided, and participants only listened to the audio while exhaling. The visual scene factor included three levels: “multi-effects,” where the blackboard vibrated, broke, and cracks appeared in the window shortly after the pseudo-loud vocalization began; “blackboard,” where the blackboard broke abruptly without prior indication; and “sight-off,” where no visual changes occurred in response to the pseudo-loud vocalization. These conditions were designed to investigate the impact of visual scene on vocal agency. The distinction between “multi-effects” and “blackboard” was specifically established to examine the effect of the quantitative difference in visual scene.

3.3 Subjective Ratings

Immediately following each trial, participants completed a questionnaire shown in Table 1, using a 7-point Likert scale to evaluate the sense of vocal agency. A score of 1 indicated the absence of the sensation, while a score of 7 represented the strongest sensation. To evaluate changes in participants’ emotions during the loud vocalization, the Japanese version of the Positive and Negative Affect Schedule (PANAS) [13, 33] was used. This selection is based on the position, supported by prior research [14], that stress strongly correlates with the positivity and negativity of emotions.

3.4 Procedure

Upon arrival at the laboratory, participants were provided with a detailed explanation of the experiment and signed a consent form upon agreeing to participate. Prior to the experiment, demographic and background information was collected, including gender, age, the presence of hearing impairments, the adequacy of their vision for daily activities, and their prior experience with VR.

Participants were equipped with the HMD, laryngeal device, and headphones and received a detailed explanation of the experimental flow and the graphical user interface (as shown in Figure 4). To familiarize participants with the procedure and minimize order effects, multiple practice sessions were conducted beforehand. Initially, participants observed a classroom scene and practiced following the vocalization timing displayed on the GUI while receiving only auditory stimuli. During this phase, participants were instructed to slightly open their mouths and exhale slowly while imagining themselves producing the vocalization of the vowel /a/ loudly. This step allowed participants to become familiar with the interface and timing. Subsequently, participants practiced under the “vib-on” condition multiple times to become familiar with the sense of vocal agency induced by the laryngeal vibratory stimuli. They were also briefed on the visual scene they would encounter, including the destruction of the blackboard, the vibration of the blackboard, and cracks forming in the window. Additionally, the questionnaire items were explained to participants beforehand.



Figure 4: Participants wearing the device during the experience.

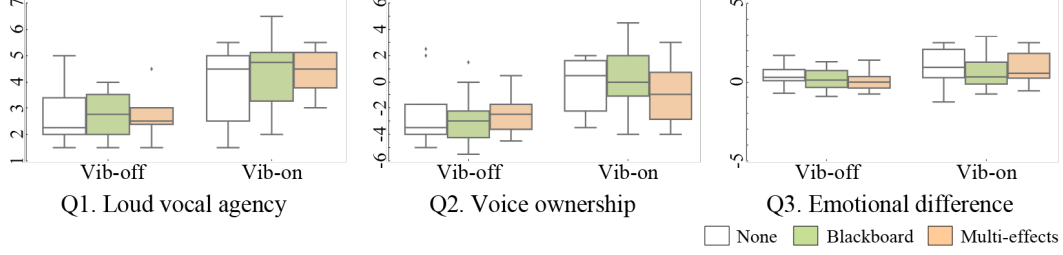
The experiment consisted of 12 trials structured as a 2-factor design with 2 levels of vibration, 3 levels of visual stimuli, and 2 repetitions. The trial order was counterbalanced using a balanced latin square design [3] to mitigate order effects. To provide participants with a more accurate reference for the sensation of actual loud vocalization, which served as a baseline for the questionnaire evaluations, tasks involving real loud vocalization were conducted before the start of the experimental trials and again after the completion of six trials.

3.5 Statistical Analysis

All questionnaire results consisted of ordinal data that were non-normally distributed. The data were summarized by calculating the median for each condition repetition. For PANAS data, the mean scores of the 10 items in the positive and negative affects were calculated separately. The negative score was then subtracted from the positive score to generate a single metric representing emotional valence. This resulted in “PA-NA difference” values ranging from

Table 1: Three questions answered regarding each trial of the experiment.

Variable names	Questionnaire statements
Loud vocal agency	I felt as though I was vocalizing loudly, similar to the pre-experiment vocalization at 80 dB.
Ownership of voice	I felt as though I was hearing my own voice.
Ownership of voice-control	I felt as though I was hearing someone else's voice.

**Figure 5: Response results for the three items: loud vocal agency, voice ownership, and emotional difference.**

-5 to 5. Following data preprocessing, an ART-ANOVA [35] was employed to account for interactions without assuming normality. Post-hoc tests using the ART-C test [5] were conducted after confirming significant interactions. All tests were conducted using Python and R with a significance level of 0.05 used for all analyses.

3.6 Participants

Eight participants were recruited via a mailing list to participate in the experiment. Participants consisted of 5 males and 3 females, with an average age of 27.5 years (SD = 4.82). Each participant received a remuneration of 1,170 JPY for their participation. The study received approval from the research ethics review committee, and informed consent was obtained from all participants prior to the commencement of the experiment.

4 Results

The questionnaire responses are summarized in Figure 5, and the ANOVA results, examining interactions between the two factors and the main effects of each factor, are presented in Table 2. As a result, no interactions were observed across all items, and only the vibratory factor showed significant main effects. Post hoc test results, as shown in Table 3, indicated that the vibratory stimuli enhanced the loud vocal agency, strengthened the voice ownership, and slightly shifted the PA-NA difference in a positive direction. Although visual scene were not significant for these measures, it is noteworthy that they were deemed significant in the evaluation of the negative affect ($F_{2,35} = 4.69, p = 0.02$).

5 Discussions

5.1 The effect of vibratory stimuli

The enhancement of loud vocal agency through vibratory stimuli, as previously reported [21], is consistent with our findings. The observed increase in voice ownership is particularly noteworthy. In this experiment, all participants, irrespective of gender, were

Table 2: ART-ANOVA results for each of the three question items.

	Loud vocal agency	Voice ownership	PA-NA difference
Vibration	$F_{1,35} = 42.1$ $\eta_p^2 = 0.55$	$F_{1,35} = 37.1$ $\eta_p^2 = 0.51$	$F_{1,35} = 8.20$ $\eta_p^2 = 0.19$
Visual scene	$F_{2,35} = 0.38$ $\eta_p^2 = 0.02$	$F_{2,35} = 0.36$ $\eta_p^2 = 0.02$	$F_{2,35} = 1.04$ $\eta_p^2 = 0.06$
Interaction	$F_{2,34} = 0.82$ $\eta_p^2 = 0.04$	$F_{2,35} = 1.71$ $\eta_p^2 = 0.09$	$F_{2,35} = 0.45$ $\eta_p^2 = 0.03$

■: $p < 0.01$, ■: $p < 0.05$, □: n.s.

Table 3: Multiple comparison results for each of the three question items.

	Loud vocal agency	Voice ownership	PA-NA difference
Vib-off	2.75	-3.00	0.125
Vib-on	4.75	-0.25	0.475

** : $p < 0.01$, * : $p < 0.05$, + : $p < 0.10$

exposed to the same audio stimuli, which likely introduced a perceived discrepancy between their own voice and the presented sound. Among three female participants, one reported minimal discomfort with the male voice used in the experiment, while another expressed significant discomfort. Nevertheless, vibratory stimuli generally enhanced the voice ownership, elevating the auditory stimuli—initially perceived as someone else's voice—to a level where participants' judgments became ambiguous. This trend was observed across genders, suggesting a degree of flexibility in voice ownership. Notably, this flexibility appears to extend beyond voices

of the same gender, as previously reported [38]. This finding can be attributed to the application of vibrations, matching the frequency and timbre of the audio, to the larynx—a key articulatory organ—prompting participants to associate unfamiliar auditory stimuli with self-generated sounds. Furthermore, the emotional effects of vibratory stimuli support the hypothesis that the loud vocalization has an arousing influence on emotional states, amplifying its positive emotional impact.

5.2 The effect of visual scene

The visual scene intended to evoke a positive emotional shift by providing participants with a sense of exhilaration showed no significant effects across most measures. This finding appears to result from participants' perceptions of the visual scene, as reported in the verbal questionnaire administered post-experiment. Notably, several participants expressed feelings of surprise, fear, or guilt regarding the destruction of objects depicted in the scene. This individual variation may be related to participants' tendencies in employing defense mechanisms when exposed to stress. Participants who frequently resort to acting out may be better suited to benefiting from pseudo-loud vocalization. This participants' tendency may be partially age-dependent, as suggested by Whitty's findings that individuals aged 17–23 years often rely on immature defense mechanisms [34]. Additionally, the effectiveness of destruction-themed content in stress relief may depend on participants' prior exposure to similar themes in media such as manga and anime. If an individual has previously encountered narratives in which destruction is depicted as a means of stress relief, they may be more likely to respond positively to such content. Future research should aim to differentiate individuals who perceive destructive visual scenes as exhilarating from those who experience fear or discomfort, enabling a more targeted analysis of their respective responses.

This study hypothesized that the delayed onset of visual stimuli following vocalization would enhance the loud vocal agency. Contrary to expectations, this effect was not observed. This result may be attributed to the lack of intuitive understanding of how the invisible act of vocalization interacts with the external environment, resulting in weak self-attribution of the visual effects. A more effective approach might involve creating clearer associations between vocalization and its effects, such as synchronizing vocalization with the release of tangible objects by an avatar in the VR environment.

6 Limitations and Future work

The experimental setup in this study imposed constraints on the exhilaration for the sake of experimental control, potentially limiting the full realization of the potential benefits of pseudo-loud vocalization. This limitation may have negatively impacted the results. Additionally, the small sample size, the absence of physiological measures, and the lack of comparison with existing methods restrict the study's contributions and represent key areas for future improvement. Furthermore, the long-term effects of pseudo-loud vocalization remain unexplored. Future research should focus on categorizing users according to their anticipated behaviors under stress and developing applications of pseudo-loud vocalization to suit each user group.

7 Conclusion

This study introduced a stress-relief method utilizing pseudo-loud vocalization in VR to neutralize “acting out,” a known manifestation of problematic behaviors under stress. To enhance the stress-relief effect, a safe, destructive VR experience was incorporated. Findings from the user experiments demonstrated that laryngeal vibratory stimuli significantly enhanced the loud vocal agency, thereby amplifying the positive emotional effects of vocalization. Conversely, destructive visual scene failed to exhibit a general stress-relief effect and instead elicited feelings of surprise or fear among participants.

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